CCC-C-428E
November 5, 1971
SUPERSEDING
Fed. Spec. CCC-C-428D
August 5, 1965

FEDERAL SPECIFICATION

CLOTH, DUCK, COTTON; FIRE, WATER, WEATHER, AND MILDEW RESISTANT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for fire, water, weather, and mildew resistant cotton duck.

1.2 Classification.

1.2.1 Types and classes. The finished cloth shall be of the following types and classes as specified (see 6.2):

Type I - Cloth finished in Olive Drab No. 7 color.
Type II - Cloth finished in deck grey color.
Class 1 - Regular finish.
Class 2 - Dry finish.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

CCC-C-419 - Cloth, Duck, Cotton, Unbleached, Plied-Yarns, (Army and Numbered).
CCC-C-443 - Cloth, Duck, Cotton; (Single and Plied Filling Yarns Flat).
PPP-P-1135 - Packaging and Packing of Duck Fabrics (Cotton; Synthetic Fiber: Cotton-Synthetic Fiber Blends).

Federal Standards:

Fed. Std. No. 4 - Glossary of Fabric Imperfections.
Fed. Std. No. 191 - Textile Test Methods.

FSC 8305

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices, in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C., 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is established, the cloth shall match the standard sample for shade and shall be equal to, or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.2 Laboratory report approval and bid sample approval. Unless otherwise specified at the time of submission of a bid, the prospective supplier shall

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submit to the contracting officer for approval, a sample 5 by 9 inches (minimum) and a certified copy of a recent laboratory report covering the type (i.e. flat duck, numbered duck, etc.) which he proposes to deliver (see 6.2). Unless otherwise specified by the contracting officer, the bidder shall certify that the sample and laboratory report submitted cover cloth which was finished in the mill where the finishing will be performed if the contract is awarded. The laboratory report shall contain test data which demonstrate that the finished cloth, which the supplier proposes to deliver, has been tested and has been found to comply with all requirements of this specification. Any of the following types of reports will be satisfactory from the standpoint of this requirement:

(a) An independent or commercial laboratory report.
(b) The prospective bidder's own laboratory report.
(c) A Government laboratory report from a contract within 6 months of date of submission of bid.

The purpose of the above requirements is to assist the Government to determine the capability of bidders to manufacture a cloth meeting all the requirements of the specification. The submission of an acceptable sample under this requirement shall not be construed as relieving a supplier from subsequently meeting all requirements of the specification on all deliveries.

3.3 Formula approval. The prospective supplier or his finishing plant shall submit with their bid to the contracting officer for approval, a list of materials which the prospective supplier or his finishing plant intends to use, stipulating grades and the standards of all materials including solvents and manufacturer's name. This list is for the information of the Government only, and is not to be made available to the public (see 6.5). The use of compounds containing mercury in any form are prohibited. Approval of the qualitative formula submitted shall in no way absolve the supplier from complying with the performance requirements of this specification.

3.4 Materials.

3.4.1 Basic cloth. The cloth shall be as specified by the contracting agency and shall conform to CCC-C-419 or CCC-C-443 (see 6.2), except that the nonfibrous material for CCC-C-443 shall be as specified in 3.4.1.1.

3.4.1.1 Nonfibrous material content. The cloth prior to finishing, shall contain not more than 2.5 percent starch and protein content excluding chloroform-soluble and water-soluble material and not more than 5.0 percent starch and protein content including chloroform-soluble and water-soluble material when tested as specified in 4.4.

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3.4.2 Grey-cloth weight. Unless otherwise specified in subsequent tables and paragraphs, where reference is made to grey-cloth weight, the minimum acceptable grey-cloth weight as given in the applicable grey-cloth specification shall be used to determine the requirements under these paragraphs.

3.4.3 Compound. The compound to provide fire, water, weather, and mildew resistance shall be well ground and evenly blended. Compounds containing sulfur may be permitted when specifically indicated by name in formulas approved by the contracting officer. The compound shall be of the same ingredients as contained in the list of materials submitted and approved under 3.3.

3.5 Finished cloth.

3.5.1 Construction. The number of yarns per inch of the finished cloth shall conform to the requirements of the specification covering the grey cloth specified except that a minus tolerance shall apply to the number of filling yarns, as follows:

Designation	Minus tolerance
Flat ducks (requiring desizing)	Up to 10 percent
Army duck	Up to 5 percent
No. 8 and lighter duck	Up to 5 percent
Heavier than No. 8 duck	Up to 3 percent

3.5.2 Finished width. The nominal over-all width of the finished cloth, inclusive of selvages, shall be as specified (see 6.2) with the following tolerances:

Numbered duck

Width of duck Inches	Tolerance Inch
40 or less	-1/2 +1/4
Over 40 to 51 inclusive	-1/2 +3/8
Over 51 to 81 inclusive	-3/4 +1/2
Over 81	-7/8 +5/8

Army duck

Width of duck Inches	Tolerance Inch
40 or less	-1/4 +1/2
Over 40 to 51 inclusive	-1/4 +5/8
Over 51 to 81 inclusive	-3/8 +3/4
Over 81	-1/2 +7/8

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3.5.3 Breaking strength. The breaking strength of the finished cloth in both warp and filling directions shall be not less than that of the corresponding specification requirements for the grey cloth.

3.5.4 Weight. The weight of the finished cloth shall be the actual weight of the grey cloth plus an allowance for the treatment in accordance with table I.

TABLE I

Grey weight (ounces per sq. yd.)	Allowable maximum increase in weight due to treatment based on actual grey-cloth weight (percent)
12 and over	50
Under 12	65

3.5.5 Color. Unless otherwise specified, the color shall be as specified for the applicable type and shall match the standard sample when furnished by the contracting activity (see 6.4). The color shall be obtained with carbon black or lamp black, inorganic or phthalocyanine pigments, or such other coloring materials as shall not be appreciably affected by exposure to weathering when tested as specified in 4.4.

3.5.5.1 Colorfastness. When a standard sample is established, the colored and finished cloth shall show colorfastness to accelerated weathering equal to or better than the standard sample when tested as specified in 4.4. When a standard sample is not available or is not referenced as a standard for colorfastness, the colored and finished cloth shall show "good" colorfastness to accelerated weathering when tested as specified in 4.4.

3.5.5.2 Matching. The color of the finished cloth shall match the standard shade under a natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.5.6 Crocking.

3.5.6.1 Class 1. Cloths with the regular finish shall show no more crocking on dry white cotton cloth than the Munsell value of 5.0 when tested as specified in 4.4.

3.5.6.2 Class 2. Cloths finished with a dry finish for upholstery purposes shall show no more crocking on dry white cotton cloth than Munsell value of 7.0 when tested as specified in 4.4.

3.5.7 Flexibility. The finished cloth shall show a bending moment of not more than that specified in table II when tested as specified in 4.4.

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TABLE II. Flexibility

CCC-C-419 cloth designations	Initial bending moment inch-lbs. maximum	After heating at 200°-205°F. (bending moment inch-lbs.) maximum	At 0° ± 5°F. (bending moment inch-lbs.) maximum
Numbered			
2	0.16	0.400	0.400
4	0.12	0.300	0.300
6	0.06	0.180	0.180
8	0.05	0.125	0.125
10	0.03	0.075	0.075
12	0.02	0.050	0.050
Army ducks			
8.25	0.013	0.032	0.032
9.85	0.013	0.032	0.032
12.29	0.013	0.032	0.032

3.5.7.1 For all other cloths, the flexibility requirements shall be as specified (see 6.2).

3.5.8 Water resistance. The finished cloth shall have a maximum leakage of 50 ml. at the height indicated in table III when tested by the water-permeability test specified in 4.4. The cloth shall conform to the same maximum leakage requirements after the accelerated weathering test specified in 4.4.

3.5.8.1 The requirements for resistance to water permeability shall be applicable to all ducks of CCC-C-419 only.

3.5.8.2 Retest. When the treated cloth fails to comply with 3.5.8, a retest shall be conducted and if the test specimens, after immersion in water at a temperature of 75° to 85°F. to a depth of 2 inches for 2 hours and tested while still wet, meet the requirements of 3.5.8, the material shall be considered as having passed the water resistant requirement.

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TABLE III. Height for test of water resistance

Untreated weight (oz./sq. yd.)	Hydrostatic height		Time of exposure, minutes
	Inches	Centimeters	
Over 20	20	50.80	10
14 thru 20	14	35.56	10
Under 14	8	20.32	10

3.5.9 Flame resistance. When tested as specified in 4.4, the time of flaming of the specimens shall not exceed 2 seconds. The average length of char shall not exceed the requirements of table IV. The cloth shall conform to the same fire-resistance requirements after the accelerated weathering period as specified in 4.4 and also after being exposed to water leaching when tested as specified in 4.4.

TABLE IV. Average length of char

Untreated weight (oz./sq. yd.)	Maximum average length of char (inches)
Under 10	4.5
10 thru 20	3.5
Over 20	2.0

3.5.10 Mildew resistance.

3.5.10.1 For type I cloths. The cloth shall be made mildew resistant by proper deposition within the cloth of a fungicide which is currently approved for this use by the procuring activity. Unless otherwise specified (see 6.2), the fungicide shall be one of the following in the specified concentration:

(a) Copper 8-quinolinolate to deposit a minimum of 0.18 percent copper as metal from copper 8-quinolinolate to a maximum of 0.27 percent copper as metal from copper 8-quinolinolate.

(b) Copper naphthenate combined with copper 8-quinolinolate to deposit a minimum of 0.35 percent total copper with a minimum of 0.05 percent copper as metal from copper 8-quinolinolate to a maximum of 1.48 percent total copper with a maximum of 0.18 percent copper as metal from copper 8-quinolinolate.

The fungicide shall be uniformly applied and the cloth shall be well penetrated.

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3.5.10.2 For type II cloths. The cloth shall be made mildew resistant by proper deposition within the cloth of a fungicide which is currently approved for this use by the procuring activity.

3.5.11 Breaking strength after accelerated weathering. The requirements for breaking strength after accelerated weathering shall be applicable for civil agencies procurements only.

3.5.11.1 Army and numbered duck. The breaking strength of the finished cloth after exposure to 100 hours accelerated weathering as specified in 4.4 shall conform to the following requirements:

Untreated (grey weight) (ounces per square yard)	Minimum percent retention of finished cloth breaking strength
Under 15 ozs.	60.0 percent
15 oz. and over	75.0 percent

3.5.11.2 Flat ducks. The breaking strength for flat ducks CCC-C-443, after accelerated weathering, shall be not less than 80 percent of the finished cloth, or not less than that specified in the base cloth specification after exposure to 200 hours accelerated weathering, when tested as specified in 4.4.

3.6 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act (see 2.2).

3.7 Length and put-up. The cloth shall be put-up on rolls in accordance with PPP-P-1135. The minimum length of pieces shall be in accordance with the applicable grey cloth specification. The maximum length of rolls, when converted to pounds shall not exceed the maximum weight specified in PPP-P-1135.

3.8 Face identification. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the roll.

3.9 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement document.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with the provisions of 4.2.2.1 through 4.2.2.3.

4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in Section I of Fed. Std. No. 4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 with the following exceptions: glossary numbers 1, 2, 3, 5, 14, 26, 27 and 29 shall not be scored unless they exceed three times the thickness of the yarn. Misdraws and reed marks shall be scored if they result in a clearly noticeable separation of the warp yarns. For cloth lighter than No. 10 duck, only those knots and slubs which exceed the limits shown on figure 1 of Fed. Std. No. 4, shall be scored as defects. For No. 10 duck and heavier, knots and slubs shall be scored as a defect when it exceeds three times the normal yarn size. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls. The lot shall be unacceptable if the points per

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100 square yards of the total yardage examined exceeds 40 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual roll exceeds 60 points. If one roll exceeds 60 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 60 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects when present shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth.
Color not uniform, off shade, (not within established tolerance), shaded end to end, side to side, or side to center.
Poor dye penetration, mottled, streaky or cloudy.
Overall application of compound not uniform.
Objectionable odor.
Overall uncleanness.
Clearly visible crystallization of mildew inhibitors.
Poorly constructed - not firmly or tightly woven.
Bias or bowed filling.
Tackiness (sticky to the touch).
Width not within specified tolerance.

4.2.2.2 Examination for length. During the yard-by-yard examination, each roll in the sample shall be examined for put-up and length in accordance with the provisions of the untreated cloth specification.

4.2.2.3 Examination for compliance with Textile Fiber Products Identification Act and Face Marking.- During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

Not labeled, ticketed or invoiced in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

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Face marking missing from either or both ends.

Face marking on wrong side.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1135 to determine that packaging, packing, and marking complies with section 5 requirements.

4.4 Testing of the end item. The method of testing specified in Fed. Std. No. 191, whever applicable and as listed in table VI shall be followed. The physical and chemical values specified in section 3 (except where otherwise indicated) apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit for testing shall be 1/4 yard of the desized untreated cloth and 5 continuous yards of the finished cloth, all full width of the material. The sample size (number of sample units) shall be in accordance with table V except for the inhibitor content. The sample unit for testing the inhibitor content shall be 1/4 yard of the finished cloth and the sample size shall be 5 for all lots less than 22,001 units in size, and 8 for all lots over 22,000 units in size. The lot size shall be expressed in units of 1 yard each. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified, or if the lot average determination for allowable increase in weight due to treatment exceeds requirements. Unless otherwise specified, all physical tests shall be performed under standard conditions as defined in Fed. Std. No. 191. When the data in the "number of determinations per individual sample unit" and "results reported as" columns are not specified in table VI, they shall be as required by the referenced test method. All test reports shall contain the individual values utilized in expressing the final results.

TABLE V. Sample size

Lot size in yards	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE VI. Test methods

Characteristic	Requirement paragraph	Test method	Number of determinations per individual sample unit	Results reported as
Presence of mercury	3.3	6/	-	-
Nonfibrous material content	3.4.1.1	2611	-	-
Materials:				
Compound	3.4.3	6/	-	-
Finished cloth:				
Weight	3.5.4	5041	5	Average of 5 determinations to nearest 0.01 ounce. 7/
Yarns per inch	3.5.1	5050	-	Pass or fail.
Breaking strength:				
Initial	3.5.3	5100	-	-
After accelerated weathering	3.5.11.1 3.5.11.2	5100, 4.4.2 1/	-	8/
Flexibility:				
Initial				
Warp	3.5.7	5202, 4.4.1	4	Average of 4 determinations to nearest 0.001 inch pound.
Filling	3.5.7	5202, 4.4.1	4	Average of 4 determinations to nearest 0.001 inch pound.

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TABLE VI. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method	Number of determinations per individual sample unit	Results reported as
After heating aging				
Warp	3.5.7	5202, 4.4.1	4	Average of 4 determinations to nearest 0.001 inch pound.
Filling	3.5.7	5202, 4.4.1	4	Average of 4 determinations to nearest 0.001 inch pound.
At low temperature				
Warp	3.5.7	5202, 4.4.1	4	Average of 4 determinations to nearest 0.001 inch pound.
Filling	3.5.7	5202, 4.4.1	4	Average of 4 determinations to nearest 0.001 inch pound.
Colorfastness to weathering	3.5.5.1	5671 1/ 2/	-	-
Fastness to crocking	3.5.6	5651	-	-
Flame resistance:				
Initial				
Char length				
Warp	3.5.9	5903	4	Average of 4 determinations to nearest 0.1 inch.
Filling	3.5.9	5903	4	Average of 4 determinations to nearest 0.1 inch.

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TABLE VI. Test method (cont'd)

Characteristic	Requirement paragraph	Test method	Number of determinations per individual sample unit	Results reported as
Flame resistance: (cont'd)				
Initial (cont'd)				
Flaming time				
Warp	3.5.9	5903	4	Each determination and average of 4 determinations to nearest 1/5 second.
Filling	3.5.9	5903	4	Each determination and average of 4 determinations to nearest 1/5 second.
After accelerated weathering:				
Char length				
Warp	3.5.9	5903, 4.4.2 1/	4	Average of 4 determinations to nearest 0.1 inch.
Filling	3.5.9	5903, 4.4.2 1/	4	Average of 4 determinations to nearest 0.1 inch.
Flaming time				
Warp	3.5.9	5903, 4.4.2 1/	4	Each determination and average of 4 determinations to nearest 1/5 second.
Filling	3.5.9	5903, 4.4.2 1/	4	Each determination and average of 4 determinations to nearest 1/5 second.

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TABLE VI. Test method (cont'd)

Characteristic	Requirement paragraph	Test method	Number of determinations per individual sample unit	Results reported as
After water leaching:				
Char length				
Warp	3.5.9	5903, 5832 3/	4	Average of 4 determinations to nearest 0.1 inch.
Filling	3.5.9	5903, 5832 3/	4	Average of 4 determinations to nearest 0.1 inch.
Flaming time				
Warp	3.5.9	5903, 5832 3/	4	Each determination and average of 4 determinations to nearest 1/5 second.
Filling	3.5.9	5903, 5832 3/	4	Each determination and average of 4 determinations to nearest 1/5 second.
Dye identification	3.5.5	6/	-	-
Water resistance:				
Initial	3.5.8	5516	-	-
After accelerated weathering	3.5.8	5516, 4.4.2 1/	3	Average of 3 determinations to nearest 1.0 cc.
Mildew resistance				
Type I:				
Total copper analysis	3.5.10.1	2050 or 2051 4/	-	-
Fungicide identification and concentration	3.5.10.1	5/	-	-
Type II:				
Fungicide	3.5.10.2	6/	-	-

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1/ The time of exposure shall be 100 standard hours.

2/ The test apparatus shall be as specified in 4.4.2, except that the filters shall not be removed.

3/ Leaching period shall be 24 hours.

4/ Suppliers shall indicate on their test reports which fungicides were used.

5/ The supplier shall certify that the required amount of copper as metal from copper 8-quinolinolate has been used in the treatment of the duck. The certificate shall be accompanied by actual test, inspection, or other verifiable quality data.

6/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

7/ All test results on the weight of the basic untreated cloth and finished cloth shall be averaged separately to determine the respective lot average weights. These values shall then be utilized in calculating the percent increase in the finished cloth weight. Results shall be calculated to the nearest 0.1 percent.

8/ All test results shall be averaged and this value shall be utilized in calculating the percent change from the initial treated cloth value. Results shall be calculated to the nearest 0.1 percent.

4.4.1 Flexibility tests. The test shall be conducted in accordance with method 5202 with the following modifications:

Load scale reading. The load scale reading shall be taken at a 20 degree angular deflection of the specimen.

Specimens. Eight specimens, 1 inch by 2 inches, shall be cut: four with the long dimension in the warp direction, and four in the filling direction, and pressed between two 6- by 6-inch glass plates weighted with a 20-pound weight for four hours at standard condition prior to testing.

4.4.1.1 After heat aging. Flexibility test specimens shall be cut from the conditioned specimens as specified in 4.4.1. However, prior to putting them under glass plates and a 20-pound weight, the specimens shall be subjected to a temperature between 200° to 205° F. for 120 hours in a well ventilated oven and then allowed to cool at standard conditions for 16 hours. The heat aged and conditioned samples shall then be placed on glass plates under weight for four hours prior to testing.

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4.4.1.2 At low temperature. Flexibility test specimens shall be prepared as specified in 4.4.1. After the required pressing for four hours, weight and upper glass plate shall be removed. The lower plate holding the conditioned and pressed specimens shall then be subjected to a temperature of 0° ± 5° F. for not less than one hour and then tested at that temperature.

4.4.2 Accelerated weathering test.

4.4.2.1 Apparatus and procedure. The apparatus shall be in accordance with method 5804 except as modified below (see 6.3).

4.4.2.2 Filters. The filters shall be removed when testing all types of Army and numbered ducks listed in CCC-C-419 and flat ducks of CCC-C-443.

4.4.2.3 Procedure. Two samples 9-1/2 by 28 inches, constituting the material required to complete all weathering tests, shall be cut from the sample unit. One sample shall be cut with the long dimension in the direction of the warp; the other with the long dimension in the direction of the filling. When the width of the sample unit does not permit a full 28-inch specimen to be taken in the filling direction, two 9-1/2- by 14-inch samples shall be cut. The specimens shall be exposed one above the other in the quandrant of the accelerated weathering apparatus. The specimens shall be changed from the top to bottom racks approximately every 25 hours during the 100 hour exposure period (see 3.5.11). At the conclusion of the exposure period, the samples shall be removed from the apparatus and allowed to dry; then the breaking strength and fire resistance specimens shall be cut from the exposed material. All specimens shall be conditioned at standard conditions prior to testing. The four fire resistance specimens shall be cut from the exposed samples such that the direction being tested shall have been exposed to accelerated weathering in the vertical position. In conducting fire resistance tests on these specimens, two specimens shall be subjected to the test flame at the end which was at the top of the exposed material, and two shall be subjected to the test flame at the end which was at the bottom of the specimen. The lower edges of the fire resistance test specimens shall be trimmed, if necessary, so that a freshly cut end is exposed to the test flame.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

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5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1135.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1135.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1135.

6. NOTES

6.1 Intended use.

6.1.1 Types I and II.

6.1.1.1 Class 1. Cloth treated for fire, water, weather and mildew resistance as covered by this specification for class 1 is intended for use in the manufacture of canvas covers, tents, tarpaulins, and other duck items providing protection under conditions of prolonged outdoor use.

6.1.1.2 Class 2. Cloth treated for fire, water, weather, and mildew resistance as covered by this specification for class 2 is intended for use as upholstery material in motorized vehicles and uses where a higher degree of resistance to crocking is required by the condition of the end use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Base cloth required (see 3.4.1).
(c) Type and class required (see 1.2.1).
(d) Width of cloth required (see 3.5.2).
(e) When bid sample or laboratory report is not required (see 3.2).
(f) Submission of certification as to the place of finishing, when required (see 3.2).
(g) Color required if other than specified (see 3.5.5).
(h) Flexibility requirements for duck other than numbered or Army duck specified in 3.5.7.
(i) Mildew inhibitor (if other than specified in 3.5.10.1).
(j) Selection of applicable levels of put-up, packaging and packing (see 5.1 and 5.2).

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6.3 Accelerated aging. Where the accelerated aging apparatus specified in 4.4.2 is not available, other tests and requirements may be submitted provided data is submitted showing correlation with the accelerated weathering test of table VI.

6.4 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.5 Formula disapproval. For the information of the supplier, the use of proteins and their derivatives, starch, hydrophylic materials, oxidizing oils, and oxidizing resins and resin or compounds to increase hygroscopicity may lead to formula disapproval (see 3.3).

6.6 Dark Green FWWMR Duck may be finished under the requirements for type I of this specification and Duck (dark) Blue FWWMR Duck may be finished under the requirements for type II of this specification.

6.7 Approval of flame resistants. As a part of the approval testing of flame retardant finishes or fibers for tentage use, the finished cloth shall have a maximum of 2 seconds afterflame and shall not melt or drop flaming pieces when tested in accordance with method 5905 of Fed. Std. No. 191. The finished material shall retain flame resistance characteristic after 1 year of exposure to natural weathering.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - MD
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DSA-DM

Preparing activity:

Army - GL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: CCC-C-428E

CLOTH, DUCK, COTTON; FIRE, WATER, WEATHER, AND MILDEW RESISTANT

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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